MIL-F-60998A (MU)
15 March 1970
SUPERSEDES
MIL-F-60998 (MU)
21 June 1968

MILITARY SPECIFICATION

FUZE, PD, M557, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for the M557 Point Detonating Artillery Ammunition Fuze.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of
MIL-D-46256A	- Detonator, Stab, M24, Loading, Assembling and Packing
MIL-P-60412	- Packaging, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes; General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1390

Source: https://assist.dla.mil -- Downloaded: 2026-04-23T01:08Z
Check the source to verify that this is the current version before use.

DRAWINGS

ARMY

8863535 - Fuze, Point Detonating M557

PUBLICATIONS

ARMY

ET-8863535 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Fuze assembly.-The PD, M557 Fuze assemblies shall comply with all requirements specified on Drawing (Dwg) 8863535 and with all requirements specified in applicable specifications.

3.3 Jolt and jumble.-The fuze assembly shall comply with the requirements of MIL-STD-331, tests 101 and 102.

3.4 Transportation vibration.-The fuze assembly shall comply with the requirements of MIL-STD-331, test 104.

3.5 Arming and propagation test.-The loaded fuze assembly, dwg. No. 8863535, shall arm and function high order when subjected to this test. The test shall be performed in accordance with 4.2.3.3.

3.6 Proving Ground

3.6.1 Premature bursts.-There shall be no premature bursts in the gun bore or in flight before impact when tested as specified in 4.2.3.4.

3.6.2 Functioning

3.6.2.1 Superquick.-The fuzes shall cause the projectile to function high order and shall show no evidence of reverse functioning (see 6.4) when tested as specified in 4.2.3.4.

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:08Z
Check the source to verify that this is the current version before use.

3.6.2.2 Delay.-The fuze shall cause the projectile to function high order and the fuze shall show no evidence of reverse functioning (see 6.4). The fuzes tested with a target using the 90mm gun shall cause the projectile to function within a zone of 75 feet, minimum (min.), to 250 feet, maximum (max.), behind the target when tested as specified in 4.2.3.4.

3.7 Workmanship.-All parts and assemblies shall be loaded and assembled in a thorough, workmanlike manner and all manufacturing, processing, and assembling operations shall be correctly performed. All parts shall be free of burrs, sharp edges, cracks, dirt, grease, dust, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible, and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and the certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:08Z
Check the source to verify that this is the current version before use.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Fuze metal parts components of one lot interfix number from one manufacturer.

b. Delay plunger assemblies of one lot interfix number from one manufacturer.

c. M125A1 Booster assemblies of one lot interfix number from one manufacturer.

d. M24 Detonators of one lot interfix number from one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:08Z
Check the source to verify that this is the current version before use.

Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Head assembly (see dwg. 8798232 covering a detail of dwg. 8798219).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Detonator missing or inverted ..	Visual	01001
102.	Detonator cushion missing	Visual	01002
103.	Detonator retaining screw not fully seated	Visual	01003
104.	Detonator retaining screw not staked	Visual	01004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	01005

4.2.2.2 Assembly prior to assembling head assembly and bottom closing screw (see dwg. 8798219).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Delay Plunger centrifugal plunger pin in armed condition .	Visual	02001
2.	Interrupter does not completely obstruct the flash hole	Visual	02002
Major:	AQL 0.65 Percent		
101.	Foreign matter in flash hole or on delay plunger assembly	Visual	02003
102.	Flash tube stake missing	Visual	02004
103.	Foreign matter in flash tube cavity	Visual	02005
104.	Delay plunger assembly missing .	Visual	02006

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:08Z
Check the source to verify that this is the current version before use.

Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	02007

4.2.2.3 Assembly prior to assembling booster (see dwg. 8798210).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 Percent		
101.	Total length, above max.	Gage	03002
102.	Bottom closing screw not below flush, min.	Gage	03003
103.	Ogive covers any part of setting sleeve	Visual	03004
104.	Body thread damaged	Visual	03005
105.	Delay time not specified on fuze	Visual	03006
106.	Crack or puncture at ogive head stake or dimples	Visual	03007
107.	Ogive head stake missing	Visual	03008
108.	Delay plunger inverted	Magnet-Visual	03001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	03010

4.2.2.4 Booster, prior to final fuze assembly (see dwg. 8595541 covering a detail of dwg. 8863535).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Booster in an armed or partially armed condition (see 6.5)	Visual	04001
Major:	None defined.		
Minor:	None defined.		

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:08Z
Check the source to verify that this is the current version before use.

4.2.2.5 Assembly (see dwg. 8863535).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 Percent		
101.	Length from base of booster to flange, above max.	Gage	05001
102.	Separation between booster and fuze, above max.	Gage	05002
103.	Torque test failure (booster to fuze, see 4.3.4)	Gage	05003
104.	Stake missing or improper	Visual	05004
105.	Thread damaged	Visual	05005
106.	Booster thread, pitch diameter, above max.	Gage	05006
Minor:	AQL 1.5 Percent		
201.	Marking missing, misleading, or unidentifiable	Visual	05007
202.	Protective coating missing from marking	Visual	05008
203.	Firing pin closing disc missing or damaged	Visual	05009
204.	Evidence of poor workmanship (see 3.7)	Visual	05010

4.2.2.6 Screw, detonator retaining (see dwg. 8798234).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Pitch diameter of thread, min. .	Gage	06001
Minor:	AQL 1.5 Percent		
201.	Total length	Gage	06002
202.	Pitch diameter of thread, max. .	Gage	06003
203.	Evidence of poor workmanship (see 3.7)	Visual	06005

4.2.2.7 Body assembly loading (see dwg. 9234457).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:08Z
Check the source to verify that this is the current version before use.

Major:	AQL 0.40 Percent		
101.	Tape missing or does not completely cover flash hole	Visual	07001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	07002

4.2.3 Testing

4.2.3.1 Jolt and jumble (see 3.3) - Critical defects Code No. 05020 and 05021.

4.2.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, a sample of thirty (30) loaded fuze assemblies shall be selected from each lot and subjected to the jolt test specified in MIL-STD-331, test 101. After this test, the assemblies shall not be disassembled but shall be subjected to the jumble test, MIL-STD-331, Test 102. The assemblies shall then be disassembled to determine compliance with MIL-STD-331. As a minimum inspection requirement for Critical defects, no M125A1 Booster, M1-delay plunger or body interrupter shall be in an armed or partially armed condition. If any assembly fails to comply with the requirements of MIL-STD-331, the lot shall be rejected. Test equipment shall be in accordance with 4.2.4. Assemblies subjected to these tests shall not be returned to the lot.

4.2.3.1.2 After three consecutive lots have met the criteria specified in 4.2.3.1.1, a sample of ten (10) loaded fuze assemblies shall be selected from each lot for jolt and jumble tests. Except for sample size, the test procedure and acceptance criteria shall be those specified in 4.2.3.1.1.

4.2.3.2 Transportation vibration (see 3.4) - Critical defect Code No. 05022 for safety deficiencies and Major defect, Code No. 05023 for operational deficiencies.

4.2.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, a sample of thirty (30) loaded fuze assemblies shall be selected from each lot and subjected to the transportation vibration test specified in MIL-STD-331, test 104. The assemblies shall then be disassembled to determine compliance with MIL-STD-331. As a minimum inspection requirement for Critical defects, no M125A1 booster, M1-delay plunger, or body interrupter shall be in an armed or partially armed condition. As a minimum inspection requirement for Major defects, no metal parts components shall be broken. If any assembly fails to comply with the requirements of MIL-STD-331, the lot shall be rejected. Test equipment shall be in accordance with 4.2.4. Assemblies subjected to this test shall not be returned to the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:08Z
Check the source to verify that this is the current version before use.

4.2.3.2.2 After three consecutive lots have met the criteria specified in 4.2.3.2.1, a sample of ten (10) loaded fuze assemblies shall be selected from each lot for transportation vibration tests. Except for sample size, the test procedure and acceptance criteria shall be those specified in 4.2.3.2.1.

4.2.3.3 Arming and propagation test - Major defect - Arming, Code No. 05024 - Major defect - Propagation, Code No. 05025.-The test shall be performed as specified in 4.3.2 using equipment in accordance with 4.2.4.

4.2.3.3.1 Fifty loaded fuze assemblies shall be randomly selected from each lot. All assemblies shall be tested in the delay position. The lot of fuzes shall be rejected if more than two assemblies fail to function high order.

4.2.3.4 Proving ground

4.2.3.4.1 Premature burst (see 3.6.1) - Critical defect - Any premature burst shall be cause for rejection of the lot.

4.2.3.4.2 Functioning (see 3.6.2) - Major defect

4.2.3.4.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the following acceptance criteria, one hundred (100) loaded fuze assemblies shall be selected from each lot and tested as specified in 4.3.3. The acceptance criteria and the number of fuzes to be tested shall be as specified in Table I.

TABLE I

Weapon	Target	Target Distance	Sample Size Superquick	Delay
90mm Gun	Ground Impact	Convenient	20	
90mm Gun	8" Plywood Scrn	550'		10
105mm How	Ground Impact	Convenient	20	
105mm How	Ground Impact	Convenient		20
155mm How	Ground Impact	Convenient	20	
155mm How	8" Plywood Scrn	650'		10
			60	40

Acceptance Criteria

Defect: Superquick	Accept	Reject	Code No.
Premature Burst	0	1	10001
Duds	2	3	10002
Reverse Functions	4	5	10003

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:08Z
Check the source to verify that this is the current version before use.

Delay	Accept	Reject	Code No.
Premature Burst	0	1	10004
Duds	2	3	10005
Reverse Functions	5	6	10006
Projectile functions outside zone behind screen (90mm)	3	4	10007

4.2.3.4.2.2 After three consecutive lots have met the criteria of 4.2.3.4.2.1, a total of sixty-five (65) loaded fuze assemblies shall be selected from each lot and tested as specified in 4.3.3. The acceptance criteria and the number of fuzes to be tested shall be as specified in Table II.

TABLE II

Weapon	Target	Target Distance	Sample Size Superquick	Sample Size Delay
90mm Gun	Ground Impact	Convenient	10	
90mm Gun	8" Plywood Scrn	550'		5
105mm How	Ground Impact	Convenient	15	
105mm How	Ground Impact	Convenient		15
155mm How	Ground Impact	Convenient	10	
155mm How	8" Plywood Scrn	650'		10
			35	30

Acceptance Criteria

Defect: Superquick	Accept	Reject	Code No.
Premature Burst	0	1	10011
Duds	1	2	10012
Reverse Functions	3	4	10013
Delay			
Premature Burst	0	1	10014
Duds	2	3	10015
Reverse Functions	3	4	10016
Projectile functions outside zone behind screen (90mm)	2	3	10017

4.2.3.4.3 A total of ten (10) fuze assemblies shall be set "Delay" and tested for ground impact functioning on 155mm Howitzer rounds for information only. The fuzes shall be assembled to HE loaded M107 projectiles and fired zone 2 (two) charge from an M1 model 155mm Howitzer. The expected muzzle velocity is 770 feet per second. The elevation of the weapon shall be approximately 465 mils to give an expected range of 4200 meters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:08Z
Check the source to verify that this is the current version before use.

The fuze functioning of each round shall be recorded and shall be included in the ballistic firing report for the lot.

4.2.3.5 Check-test for deterioration of detonator - Major defect, Code No. 11001.-If the total elapsed time between the acceptance of any detonator lot and assembly of that lot into fuze lots exceeds two (2) years, or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator lot shall be check-tested as specified in MIL-D-46256 immediately prior to detonator assembly into fuzes. This test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembling the detonators into the fuze. The test shall be conducted at Government expense without cost to the supplier who loaded the detonator or the supplier assembling the detonator into the fuze, and shall not constitute a basis for rejection against either supplier except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. while the detonator lot is under the jurisdiction of either contractor.

4.2.4 Inspection Equipment.-Index of Inspection Equipment List Number ET-8863535 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:08Z
Check the source to verify that this is the current version before use.

4.3 Test methods and procedures

4.3.1 Jolt and jumble.-The loaded fuze assemblies shall be subjected to the jolt test in accordance with MIL-STD-331, Test 101. After completion of the jolt test, the fuzes shall be subjected to the jumble test in accordance with MIL-STD-331, Test 102. The jumble test shall be conducted utilizing a jumble protective fixture as specified in 4.2.4.

4.3.2 Arming and propagation test.-The loaded fuze assemblies less tetryl pellets shall be armed by spinning at no more than 3000 rpm in a suitable test fixture (see 4.2.4). The assemblies shall then be impact detonated by dropping on a hard surface. The assemblies shall be dropped while spinning. A high order function shall create a hole at least one half inch in diameter in a one-eighth inch thick lead disc placed beneath the booster lead charge. Assemblies which fail to function high order shall be disassembled to determine the cause of failure. The cause of failure shall be annotated on the inspection records for the lot.

4.3.3 Proving ground.-The fuzes shall be assembled into any projectile of the specified caliber for which the fuze is standard and tested in accordance with Table III.

TABLE III

	90mm Gun	105mm Howitzer	155mm Howitzer
Recommended Model	M41	M2A2	M1
Recommended Cartridge	M71A1	M1	M107
Recommended Charge	Service	Zone 7	Zone 1 (SQ & Delay) Zone 2 (INFO Delay)
Muzzle velocity (fps)	2,400	1,550	680 (Zone 1) 770 (Zone 2)
Pressure (psi)	41,900	36,400	5,400
Spin (rpm)	19,500	13,500	3,210

*NOTE 1: Fuzes tested in 155mm How Screen Delay Phase may be assembled to projectiles loaded with a double supplementary charge in lieu of HE loaded projectiles.

*NOTE 2: Weapon evaluation in the 155mm Howitzer Ground Impact SQ Phase shall be 465 mils, approximately.

*NOTE 3: Weapon elevation in any phase shall not exceed 800 mils (45 degrees).

4.3.4 Torque test.-The disassembly torque shall be applied to the applicable component while holding the adjacent mating component in such a manner as to avoid damage to either component of the assembly.

5. PREPARATION FOR DELIVERY

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:08Z
Check the source to verify that this is the current version before use.

5.1 Packaging, packing and marking.-Packaging, packing and marking shall be in accordance with the requirements and inspection procedures specified in MIL-P-60412.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5.).

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:08Z
Check the source to verify that this is the current version before use.

for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND7. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:08Z
Check the source to verify that this is the current version before use.

6.4 Reverse functioning.-Reverse functioning is defined as superquick (SQ) functioning of the fuze when set in the "Delay" position or delay functioning of the "SQ" position.

6.5 The armed condition of the M125A1 booster is indicated when the detonator is fully visible through the cover plate. (See Figure 1). The partially armed condition of the booster is indicated when the rotor is not locked by both spring-loaded detents. (See Figures 2, 3, and 4). The unarmed condition is indicated when the rotor is locked by both spring loaded detents and the detonator is not at all visible through the cover plate. (See Figure 5).

6.6 Cost of detonator deterioration check-test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the detonation check test.

6.7 Identification of international standardization agreements.-Certain provisions of Section 3, REQUIREMENTS OF this specification are the subject of international standardization agreement ABC-ARMY-STD-56. When amendment, revision, or cancellation of this specification is proposed the departmental custodians will inform their respective department standardization offices so that appropriate action may be taken respecting the international agreement concerned.

6.8 Submission of test data.-In addition to the normal distribution of records, when the fuze is proposed by the Department of the Army, a copy of all test data shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-ND7 and DJ2
Dover, New Jersey 07801

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1390-A-789

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:08Z
Check the source to verify that this is the current version before use.

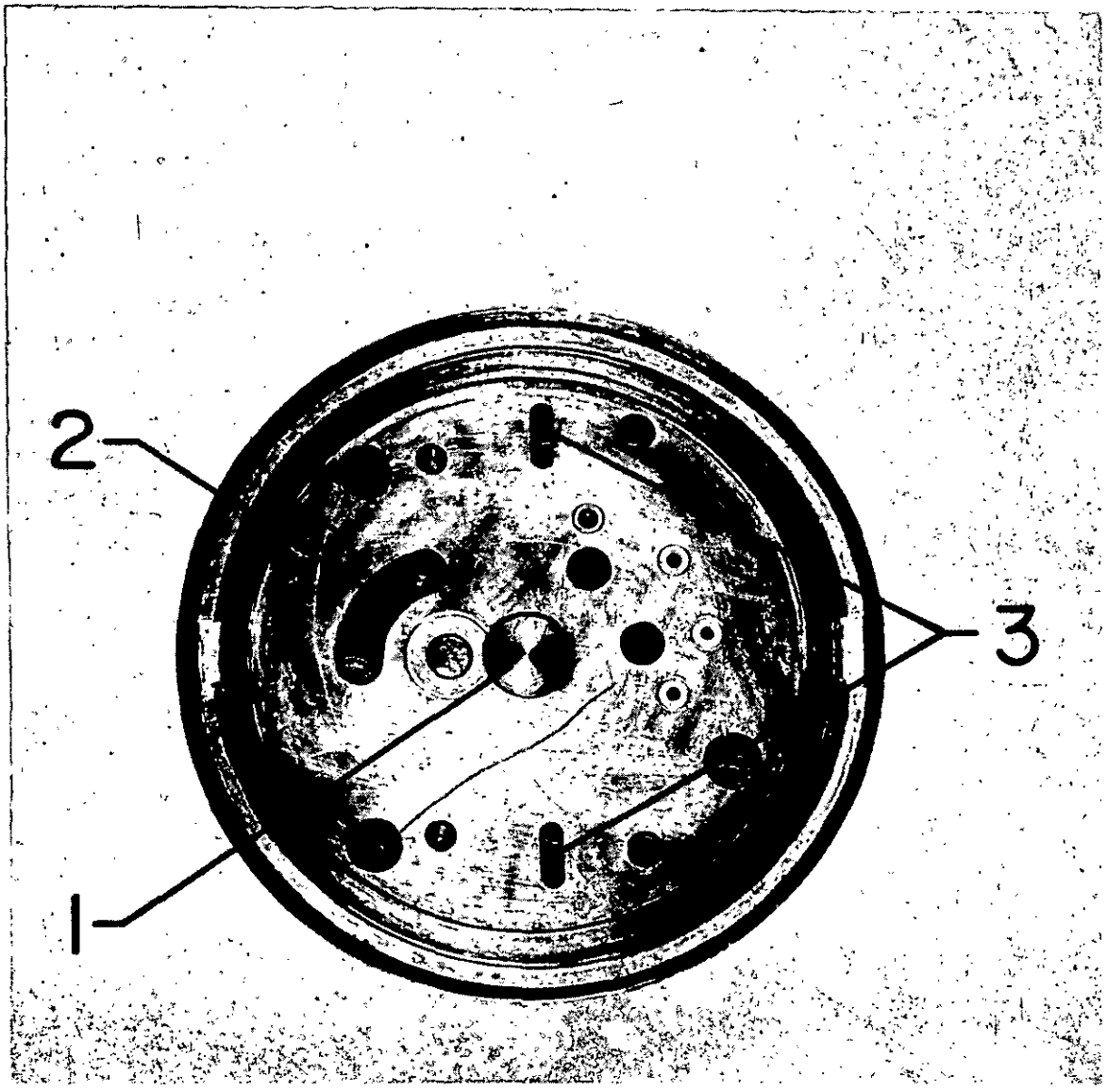


FIGURE 1.

BOOSTER ARMED
CRITICAL DEFECT

1) Red end of M17 Detonator completely visible in center of cover plate.
2) Rotor completely fills curved slot in cover plate.
3) Spring-loaded detents in IN position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:08Z
Check the source to verify that this is the current version before use.

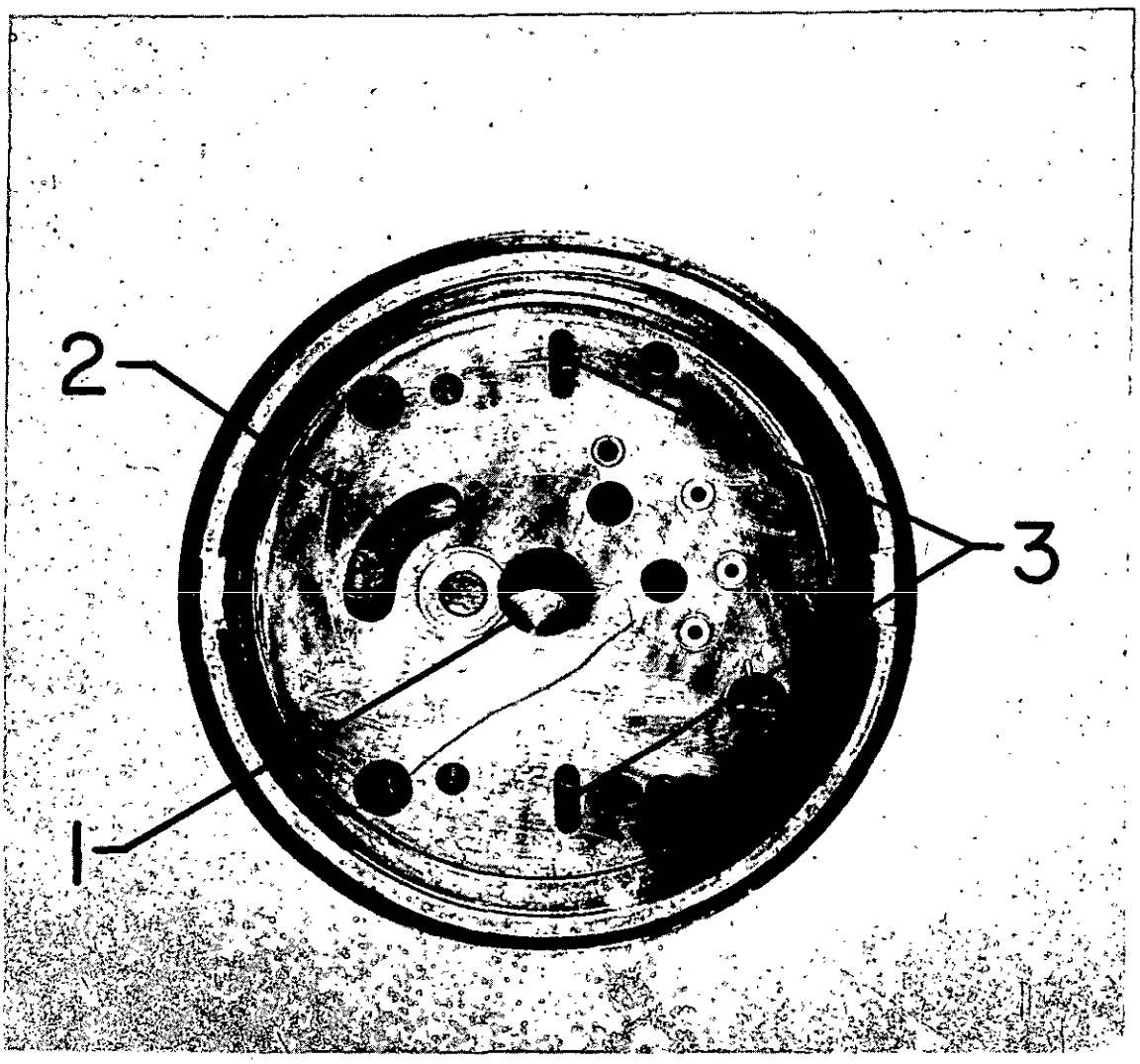


FIGURE 2.

BOOSTER PARTIALLY ARMED
CRITICAL DEFECT

1) Red end of M17 Detonator partially visible in center of cover plate.
2) Rotor completely fills curved slot in cover plate.
3) Spring-loaded detents in IN position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:08Z
Check the source to verify that this is the current version before use.

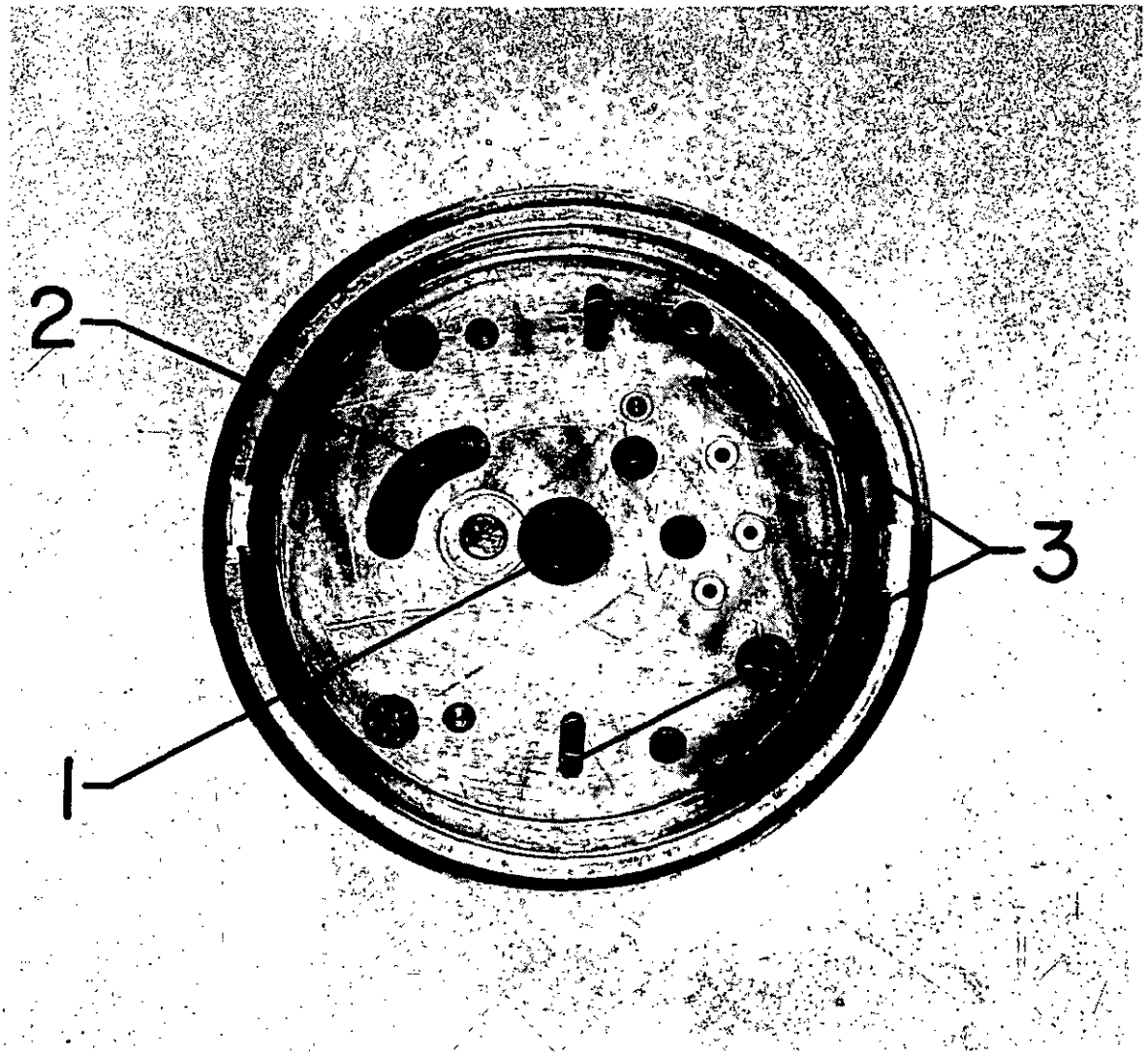


FIGURE 4.

BOOSTER PARTIALLY ARMED
CRITICAL DEFECT

1) M17 Detonator not visible in center of cover plate.
2) Rotor fills one-half of curved slot in cover plate.
3) Spring-loaded detents in OUT position.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:08Z
Check the source to verify that this is the current version before use.

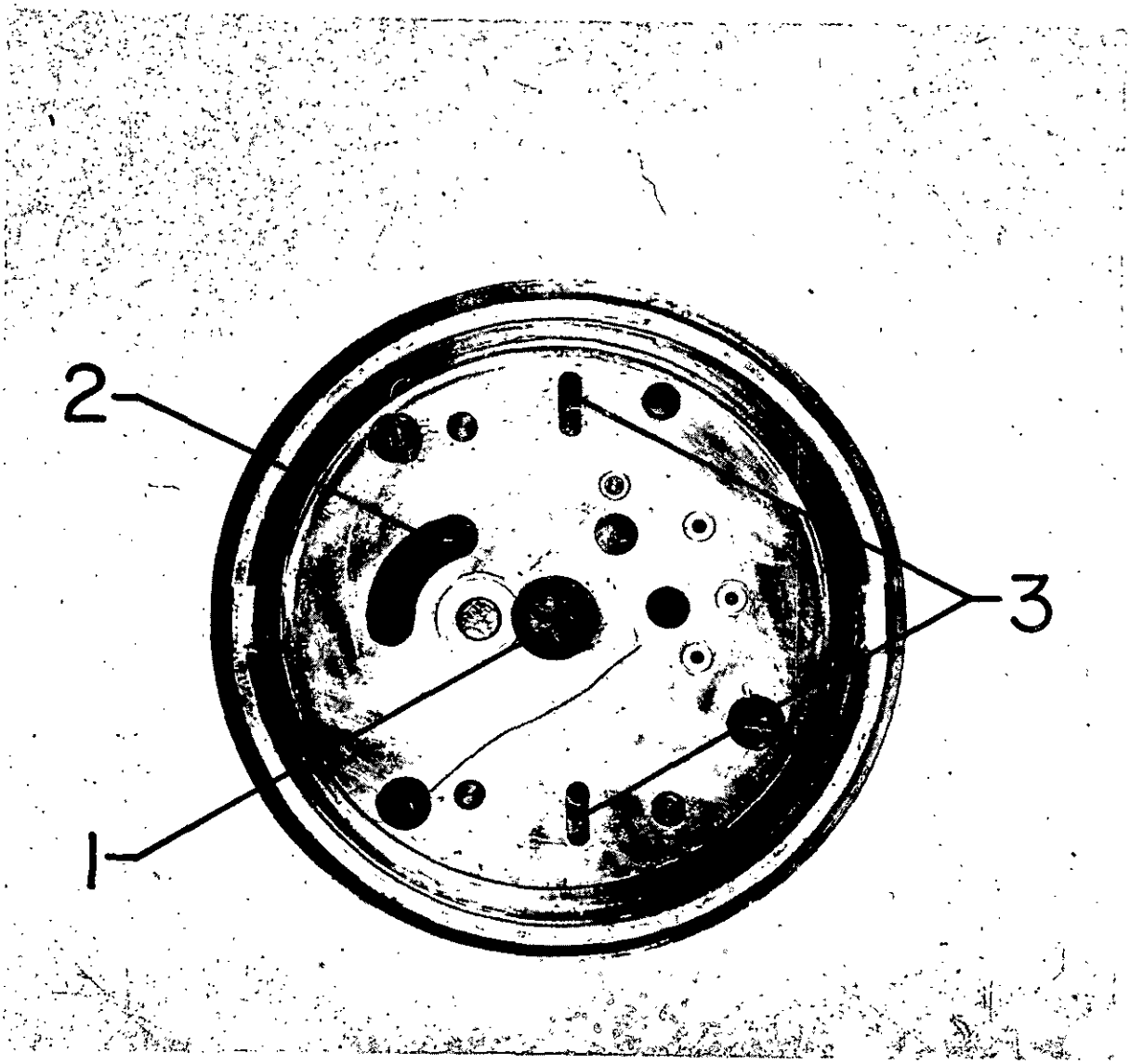


FIGURE 5.

BOOSTER NOT ARMED
SAFE CONDITION

1) M17 Detonator not visible in center of cover plate.
2) Rotor fills less than half of curved slot in cover plate.
3) Spring-loaded detents locked in IN position.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:08Z
Check the source to verify that this is the current version before use.